

TUM SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master Thesis in Computational Science and Engineering

No-Wall Boundary condition for Stellarators

Samet Kocbay



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FÜR PLASMAPHYSIK



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Author:	Samet Kocbay
Supervisor:	Prof. Dr. Sibylle Günter
Advisor:	Dr. Matthias Hölzl
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FÜR PLASMAPHYSIK



Abstract

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1 Introduction

This thesis focuses on the development of a no-wall boundary condition for non-linear magnetohydrodynamic (MHD) solvers applied to stellarator geometries. Stellarators are non-axisymmetric magnetic confinement devices currently under extensive research for the development of viable fusion reactors. By eliminating the need for large, internally driven plasma currents to generate the confining magnetic field, stellarators offer a highly stable, steady-state alternative to other confinement concepts.

The current non-linear solver relies on the assumption of perfectly conducting wall. This work removes the restriction by deriving a boundary condition expressed directly in terms of the magnetic field.

To achieve this, the research is guided by three core objectives. First, it investigates how the weak formulation of the current equation for a set of reduced MHD equations can be modified to express the boundary constraint strictly in terms of the magnetic field. Second, it explores the application of the Virtual Casing Principle to decouple the interior plasma domain from the exterior conductors, which is necessary to enable the numerical coupling between JOREK, a non-linear MHD code, and CARIDDI, a magnetostatic solver. Finally, the thesis seeks to verify whether the assembled linear response matrix accurately computes the equivalent surface currents and the resulting projected boundary magnetic field.

1.1 Magnetic Confinement and Stellarators

1.2 Non-linear MHD Simulations

1.3 The Boundary Condition

1.4 Outline

2 Theoretical Background

2.1 Magnetic Confinement and Stellarators

2.1.1 The Magnetic Confinement Principle

The aim of magnetic confinement fusion is to hold a plasma at a sufficiently high temperature and density for long enough that the energy released by fusion reactions exceeds the energy needed to sustain it [13]. A quantitative measure of how close a device comes to this goal, the Lawson triple product, is returned to when the two confinement concepts are compared in Section 2.1.2; here we are concerned only with the underlying physical principle that makes confinement possible at all.

A plasma is a quasi-neutral, ionised gas whose dynamics are dominated by collective electromagnetic interactions rather than by binary collisions [3]. Because its constituents are charged, every particle is subject to the Lorentz force

$$\mathbf{F} = q (\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (2.1)$$

where q is the particle charge, $\mathbf{v}$ its velocity, and $\mathbf{E}$, $\mathbf{B}$ the electric and magnetic fields. In a strong magnetic field the perpendicular part of Equation (2.1) forces each particle onto a tight circular gyro-orbit of radius

$$r_L = \frac{m v_{\perp}}{|q| B}, \quad (2.2)$$

the Larmor radius, while motion *along* the field line remains free (Figure 2.1). Charged particles are therefore tied to the field lines they sit on: transport across the field is suppressed to the scale of r_L , whereas it is essentially unimpeded along the field [13]. Confining the plasma thus reduces to the geometric problem of arranging magnetic field lines so that they never leave a bounded region.

The simplest bounded configuration, a straight solenoid (a magnetic mirror), still loses particles rapidly through its open ends, since nothing stops the free parallel streaming [2]. Closing the field lines onto themselves by bending the device into a *torus* removes the ends entirely, and this toroidal geometry underlies all mainline confinement concepts.

[Figure placeholder] Single-particle motion: a helical gyro-orbit around a magnetic field line, showing free parallel motion $v_{\parallel}$ and perpendicular gyration $v_{\perp}$ on the Larmor scale r_L .

Figure 2.1: A charged particle streams freely along a magnetic field line ($v_{\parallel}$) while gyrating around it on the Larmor scale r_L ($v_{\perp}$). Cross-field transport is suppressed to $\mathcal{O}(r_L)$, so the particle is effectively bound to its field line.

A purely toroidal field, however, does not confine. Such a field is necessarily stronger on the inboard side than the outboard side, decaying as $1/R$ with major radius R . The resulting field gradient and curvature drive a guiding-centre drift

$$v_D \propto \frac{\mathbf{B} \times \nabla B}{B^3}, \quad (2.3)$$

whose direction depends on the sign of the charge [15]. Ions and electrons therefore drift apart vertically, building up a charge separation and hence a vertical electric field E . The associated $E \times B$ drift is independent of charge and points radially outward, expelling the entire plasma column (Figure 2.2, left). A toroidal field alone is thus unable to hold a plasma.

The remedy is to add a *poloidal* field component, so that the total field lines wind helically around the torus rather than closing on a single poloidal plane (Figure 2.2, right). As a field line is followed once around the torus it passes alternately above and below the midplane; a particle streaming along it samples both signs of the vertical drift, which short-circuits the charge separation before it can grow. When the device is well designed, these helical field lines trace out a set of *nested toroidal flux surfaces*, the geometric backbone of a confined equilibrium that is examined in Section 2.1.3 [1].

The amount of poloidal twist is quantified by the *rotational transform* $\iota/2\pi$, defined as the average number of poloidal turns a magnetic field line completes per toroidal transit of the torus. Its inverse, the *safety factor*

$$q = \frac{2\pi}{\iota}, \quad (2.4)$$

is the equivalent measure used in tokamak physics [17]. Surfaces on which q is rational, so that field lines close on themselves after a finite number of transits, play a decisive role in magnetohydrodynamic stability and are revisited in Section 2.3.

[Figure placeholder] Left: a purely toroidal field, with the charge-dependent ∇B /curvature drift separating ions and electrons into vertical charge layers, a vertical E , and an outward $E \times B$ drift. Right: the same torus with an added poloidal twist that short-circuits the charge separation.

Figure 2.2: *Left*: in a purely toroidal field the charge-dependent ∇B /curvature drift separates ions and electrons, creating a vertical E whose $E \times B$ drift pushes the plasma outward. *Right*: adding a poloidal field component twists the field lines helically, so a particle samples both signs of the drift and the charge separation is short-circuited.

Two routes exist to generate the required poloidal field, and the choice between them distinguishes the two mainline toroidal confinement concepts: a toroidal electric current driven through the plasma itself, as in a tokamak, or a set of external, non-planar field coils, as in a stellarator [16, 7]. The consequences of this choice for the magnetic geometry, and the reason it matters for the present work, are the subject of the following section.

2.1.2 Tokamak versus Stellarator

The previous section established that confinement requires a poloidal twist of the magnetic field. The two mainline toroidal devices differ in *how* that twist is produced, and this single design choice has far-reaching consequences for the geometry, the stability, and the way the magnetic field is represented in the present work.

In a *tokamak*, the poloidal field is generated by a large toroidal electric current I_p driven through the plasma itself, typically induced by a central transformer. Combined with the toroidal field of a set of planar coils, this produces a configuration that is, to a good approximation, axisymmetric: the equilibrium is invariant under rotation about the machine axis. Axisymmetry makes tokamaks comparatively simple to design and build, and they remain the best-performing confinement devices to date. The price is the plasma current itself: it stores free energy that can drive macroscopic, current-driven magnetohydrodynamic (MHD) instabilities, and in the worst case a *disruption* — a sudden, complete loss of confinement that deposits the thermal and magnetic energy of the discharge onto the surrounding structures [17, 8]. Avoiding and controlling such

[Figure placeholder] Side-by-side comparison of a tokamak and a stellarator. Left: the (near-)axisymmetric tokamak, with planar toroidal-field coils and a toroidal plasma current I_p providing the poloidal field. Right: a classical stellarator, whose rotational transform is produced by three-dimensional, non-planar coils, requiring no net plasma current.

Figure 2.3: Schematic comparison of the two mainline toroidal confinement concepts. The tokamak relies on a driven plasma current and is approximately axisymmetric; the stellarator generates its confining field from external three-dimensional coils and is essentially current-free.

events is one of the central challenges for the tokamak line.

In a *stellarator*, by contrast, the rotational transform is supplied almost entirely by external, non-planar field coils, which generate the confining field in vacuum, independently of the plasma [16]. A stellarator therefore requires little or no net toroidal current and can operate in true steady state. Because the magnetic “cage” does not depend on the plasma state, stellarators are largely immune to the current-driven disruptions that threaten tokamaks [7]. The cost of this robustness is geometric: the field and the coils that produce it are fully three-dimensional, and the convenient axisymmetry of the tokamak is lost (Figure 2.3).

This three-dimensionality is not free of consequences for confinement. Early “classical” stellarators suffered strong neoclassical transport, because a generic 3D magnetic field confines particle guiding centres poorly, and their measured performance lagged well behind contemporary tokamaks. The modern stellarator programme overcomes this through *optimisation*: the magnetic field is shaped so that good confinement is recovered despite the lack of axisymmetry. Wendelstein 7-X (W7-X), the largest operating stellarator, was designed to approach *quasi-isodynamicity*, in which the contours of the field strength $|B|$ close poloidally [14]. An alternative family targets *quasi-symmetry*, where $|B|$ depends on only two of the three Boozer coordinates [1]; the field is then effectively symmetric as far as particle orbits are concerned, giving rise to quasi-axisymmetric (QA) and quasi-helical configurations with confinement approaching that of tokamaks [10].

The progress of these devices is measured by the Lawson *triple product* $n_i T_i \tau_E$ — the product of ion density, ion temperature, and energy confinement time — which must

[Figure placeholder] Comparison of the Lawson triple product $n_i T_i \tau_E$ achieved by representative tokamaks and stellarators, illustrating that the leading values are reached by large tokamaks while optimised stellarators such as W7-X have closed much of the gap.

Figure 2.4: Performance of tokamaks and stellarators measured by the Lawson triple product. Tokamaks currently lead, but optimised stellarators are approaching comparable values.

exceed a threshold value for a net energy-producing reactor [11]. The highest triple products achieved to date come from the largest tokamaks, and the ITER device is designed to reach reactor-relevant values; optimised stellarators such as W7-X have, however, closed much of the historical gap (Figure 2.4) [14].

These differences set the stage for the present work. Because a stellarator’s confining field is imposed externally and the device carries essentially no driven current, its non-linear MHD model can be formulated without a large toroidal plasma current and must instead resolve the field in full three-dimensional geometry, rather than reducing it by axisymmetry; this is the setting of the JOREK stellarator model used throughout this thesis. The qualification is important, however: quasi-axisymmetric configurations retain a substantial self-generated *bootstrap* current, which can reintroduce current-driven MHD activity comparable to that of tokamaks [7]. It is precisely this regime — nominally stable stellarators that nonetheless carry enough current to become MHD-active — that motivates non-linear stellarator modelling, and ultimately the free-boundary extension developed in this work. The geometric language needed to make these statements precise, in terms of nested flux surfaces and the coordinates used by the code, is introduced in Section 2.1.3, and the underlying fluid model in Section 2.2.

2.1.3 Magnetic Geometry and Topology

The well-confined configurations of the previous sections are organised around a particular magnetic geometry. This section makes that geometry precise, since the coordinates and scalar functions introduced here are exactly those used to represent the magnetic field in the remainder of the thesis.

A *magnetic flux surface* is a toroidal surface to which the magnetic field is everywhere

[Figure placeholder] Nested magnetic flux surfaces. A poloidal cross-section showing the family of nested surfaces collapsing onto the magnetic axis, with a single helical field line winding on one surface, and the poloidal angle θ and toroidal angle χ indicated.

Figure 2.5: Nested magnetic flux surfaces labelled by the flux coordinate ψ , collapsing onto the magnetic axis. A field line winds helically within a surface; the poloidal angle θ and toroidal angle χ parametrise position on it.

tangent, so that no field line crosses it,

$$\mathbf{B} \cdot \mathbf{n} = 0, \quad (2.5)$$

with $\mathbf{n}$ the surface normal. In a well-confined equilibrium the plasma volume is filled by a family of *nested* flux surfaces, each enclosing the next, labelled by a radial-like coordinate ψ that is conventionally taken to be the poloidal or toroidal magnetic flux enclosed by the surface [4]. The nesting degenerates at the innermost surface to a single closed field line, the *magnetic axis*. Because the equilibrium force balance $\mathbf{J} \times \mathbf{B} = \nabla p$ implies $\mathbf{B} \cdot \nabla p = 0$, the plasma pressure — and likewise other equilibrium quantities — is constant on each flux surface; this point is developed further in Section 2.2 [5].

This nested structure invites a natural coordinate system. Each surface can be parametrised by a *poloidal angle* θ , running the short way around the torus, and a *toroidal angle* χ , running the long way; together with the surface label ψ these form a set of *flux coordinates* (ψ, θ, χ) adapted to the magnetic geometry, in which the otherwise complicated field structure is greatly simplified [4, 1]. The rotational transform introduced in Section 2.1.2 becomes, in this picture, a property of each individual surface: $\iota = \iota(\psi)$ measures the average pitch at which field lines wind on the surface ψ , and the corresponding profile $\iota(\psi)$ (or equivalently the safety-factor profile $q(\psi) = 2\pi/\iota$) characterises the configuration [17, 7].

The arithmetic nature of ι on a given surface has a decisive topological consequence. On a *rational* surface, where $\iota/2\pi$ equals a ratio of integers, a field line closes on itself after a finite number of toroidal and poloidal transits. On an *irrational* surface a single field line never closes and instead covers the surface densely, or ergodically. Rational surfaces are the locations at which small magnetic perturbations resonate with the

[Figure placeholder] Magnetic topology at a rational surface. Left: an intact nested surface. Middle: a chain of magnetic islands formed by a resonant perturbation. Right: overlapping islands producing a stochastic field region in which flux surfaces are destroyed.

Figure 2.6: Breakup of nested flux surfaces at a rational surface: an intact surface, a magnetic island chain, and a stochastic region formed by overlapping islands.

equilibrium field, and they therefore play a central role in magnetohydrodynamic stability, as discussed in Section 2.3 [1].

Where such a resonant perturbation is present, the nested surfaces need not survive. At a rational surface the field can reconnect to form a chain of *magnetic islands*, whose interior is organised around a new, displaced set of surfaces; when neighbouring island chains grow and overlap, the field becomes *stochastic* and flux surfaces are destroyed altogether, degrading confinement (Figure 2.6) [7]. In a stellarator this is of particular concern, because the externally produced three-dimensional vacuum field can already contain low-order rational surfaces and islands before any plasma is present.

This geometry directly motivates how the magnetic field is represented in the present work. Since the field lies in nested surfaces and winds about the torus, it is naturally described by a scalar flux function together with the toroidal angle, rather than by its three Cartesian components. In the JOREK stellarator model the magnetic field is written in terms of a poloidal-flux-like function Ψ and the toroidal coordinate χ introduced above; the precise representation $\mathbf{B} = \nabla\Psi \times \nabla\chi + \nabla\chi$, together with the field-aligned operators it gives rise to, is derived in Section 2.2.3, and the functions Ψ and χ recur throughout the method developed in this thesis [12].

2.2 The Magnetohydrodynamic (MHD) Model

2.2.1 Full (Resistive) MHD

2.2.2 MHD Equilibrium

2.2.3 Reduced MHD for Stellarators

2.3 MHD Stability and the Role of the Boundary

2.3.1 Ideal-MHD Stability

2.3.2 The Conducting Wall and the No-Wall Limit

2.4 Finite-Element Discretization in JOEREK

2.4.1 Weak Formulation and the Galerkin Method

2.4.2 JOEREK's Spatial Representation

2.5 Free-Boundary Coupling Concepts

2.5.1 The Virtual Casing Principle

Describe the Virtual Casing principle [6]

2.5.2 Electromagnetics of the External Domain (CARIDDI)

The Magneto-Quasi-Static (MQS) Approximation

Integral Formulation and the Biot-Savart Law

Discretization via Edge (Whitney) Elements

3 Current Boundary Models and State of Research

3.1 The Perfectly Conducting Wall Boundary in JOREK

Starting of from the current equation in strong formulation, we now consider the current implementation used to solve the equation.

$$\tilde{j} = \Delta^* \Psi \quad (3.1)$$

To solve the initial condition equation for Ψ_0 , a boundary condition is required. The initial current $\tilde{j}_0$ is extracted directly from the GVEC equilibrium data and used to calculate the initial scalar potential Ψ_0 . For fixed-boundary simulations, the plasma domain is assumed to be enclosed by a perfectly conducting wall. This physical constraint dictates that the magnetic field cannot have a normal component at the boundary:

$$B = \nabla \Psi \times \nabla \chi + \nabla \chi \quad (3.2)$$

Under assumption (1) this leads to the form:

$$n \cdot [\nabla \Psi \times \nabla \chi] = -n \cdot [\nabla \chi] \quad (3.3)$$

... Skipping the mathematical and coordinate system tricks which leads us to be able to solve this equation for Ψ on the boundary, which we call from now on Ψ_b .

Having determined the boundary solution Ψ_b , the total initial scalar potential is linearly decomposed into two parts:

$$\Psi_0 = \Psi_{0,i} + \Psi_b \quad (3.4)$$

where $\Psi_{0,i}$ represents the interior solution. Because Ψ_b satisfies the nonhomogeneous Dirichlet boundary condition required by the perfectly conducting wall, the remaining unknown function $\Psi_{0,i}$ is explicitly defined to be zero at the boundary. The equation $\tilde{j}_0 = \Delta^* \Psi_0$ is then solved for the interior domain utilizing the standard JOREK solver with homogeneous Dirichlet boundary conditions.

3.2 Existing Free-Boundary and Wall-Coupling Approaches

4 Method

This work removes the restriction of a perfectly conducting boundary by deriving a boundary condition expressed directly in terms of the magnetic field. As a starting point, the weak formulation of the current equation is analyzed to identify the relevant coupling terms.

4.1 Weak Formulation of the Current Equation

4.1.1 Isolating the Boundary Constraint ($\nabla^\perp \Psi$)

We start off from the definition of the Current equation and will derive the weak formulation to analyze which terms we need to respect for a free boundary coupling [12]

$$\tilde{j} = \Delta^* \Psi \quad (4.1)$$

The equation is multiplied by a test function v scaled by B_v^2 for consistency with the currently implemented formulations and finally integrated over the plasma volume V .

$$\int_V B_v^2 \tilde{j} v \, dV = \int_V v \nabla \cdot (B_v^2 \nabla^\perp \Psi) \, dV \quad (4.2)$$

To reduce the order of the derivative on Ψ , apply the vector calculus product rule for divergence, $\nabla \cdot (f \mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot \nabla f$. Let the scalar field $f = v$ and the vector field $\mathbf{A} = B_v^2 \nabla^\perp \Psi$:

$$\nabla \cdot (v B_v^2 \nabla^\perp \Psi) = v \nabla \cdot (B_v^2 \nabla^\perp \Psi) + B_v^2 (\nabla^\perp \Psi \cdot \nabla v) \quad (4.3)$$

Rearranging the equation to isolate the term matching our integrand gives:

$$v \nabla \cdot (B_v^2 \nabla^\perp \Psi) = \nabla \cdot (v B_v^2 \nabla^\perp \Psi) - B_v^2 (\nabla^\perp \Psi \cdot \nabla v) \quad (4.4)$$

Substitute this expanded integrand back into the right side of the volume integral:

$$\int_V B_v^2 \tilde{j} v \, dV = \int_V \nabla \cdot (v B_v^2 \nabla^\perp \Psi) \, dV - \int_V B_v^2 (\nabla^\perp \Psi \cdot \nabla v) \, dV \quad (4.5)$$

Apply the Divergence Theorem, $\int_V \nabla \cdot \mathbf{F} dV = \oint_{\partial V} \mathbf{F} \cdot \mathbf{n} dS$, to the first term on the right-hand side. This transforms the volume integral of the divergence into a surface integral over the boundary ∂V :

$$\int_V \nabla \cdot (v B_v^2 \nabla^\perp \Psi) dV = \oint_{\partial V} v B_v^2 (\nabla^\perp \Psi \cdot \mathbf{n}) dS \quad (4.6)$$

Substituting the evaluated surface integral back provides the final weak formulation:

$$\int_V B_v^2 \tilde{j} v dV = - \int_V B_v^2 (\nabla^\perp \Psi \cdot \nabla v) dV + \oint_{\partial V} v B_v^2 (\nabla^\perp \Psi \cdot \mathbf{n}) dS \quad (4.7)$$

While this resulting surface integral isolates the boundary constraint using the perpendicular gradient $\nabla^\perp \Psi$, it must be reformulated in terms of the magnetic field B to enable the coupling of the interior plasma domain and an exterior solver Carididi.

4.1.2 Expressing $\nabla^\perp \Psi$ in terms of B

The magnetic field for the stellarator geometry is expressed in JOREK by

$$B = \nabla \Psi \times \nabla \chi + \nabla \chi. \quad (4.8)$$

Taking the cross product of B with $\nabla \chi$ and applying the vector triple product identity yields

$$\begin{aligned} B \times \nabla \chi &= (\nabla \Psi \times \nabla \chi) \times \nabla \chi + \underbrace{\nabla \chi \times \nabla \chi}_{=0} \\ &= -(\nabla \chi \cdot \nabla \chi) \nabla \Psi + (\nabla \chi \cdot \nabla \Psi) \nabla \chi. \end{aligned} \quad (4.9)$$

Introducing $B_v^2 := \nabla \chi \cdot \nabla \chi$, we obtain

$$B_v^{-2} (B \times \nabla \chi) = -\nabla \Psi + B_v^{-2} (\nabla \chi \cdot \nabla \Psi) \nabla \chi. \quad (4.10)$$

The right-hand side corresponds to the perpendicular gradient of Ψ , denoted by $\nabla^\perp \Psi$.

To verify consistency with the definition of $\nabla^\perp$ given in [12], we start from

$$\begin{aligned} \nabla^\perp &= \nabla - B_v^{-1} \nabla \chi \partial^\parallel, \\ \partial^\parallel &= B_v^{-1} \nabla \chi \cdot \nabla. \end{aligned} \quad (4.11)$$

Applying this operator to Ψ gives

$$\begin{aligned} \nabla^\perp \Psi &= \left(\nabla - B_v^{-1} \nabla \chi \partial^\parallel \right) \Psi \\ &= \nabla \Psi - B_v^{-1} \nabla \chi \left(B_v^{-1} \nabla \chi \cdot \nabla \Psi \right) \\ &= \nabla \Psi - B_v^{-2} (\nabla \chi \cdot \nabla \Psi) \nabla \chi. \end{aligned} \quad (4.12)$$

Comparing both expressions, we recover

$$\nabla^\perp \Psi = -B_v^{-2} (B \times \nabla \chi), \quad (4.13)$$

which confirms the representation. This substitution allows the weak formulation's boundary integral to be expressed strictly in terms of the magnetic field projection $(B \times \nabla \chi) \cdot n$. With the JOEREK boundary condition defined, we now proceed to formulate the external electromagnetic response computed by CARIDDI.

4.2 The Stellarator Coupling Scheme

The Virtual Casing Principle and Weak Form Formulation

To decouple the interior plasma domain from the exterior conductors, the Virtual Casing Principle is applied, following the formulation established in [9]. We assume an equivalent surface current k_{eq} exists on the boundary of the domain (Plasma domain) V_{in} that perfectly describes the magnetic field produced by the internal plasma sources in the outer domain.

The magnetic vector potential A_{in} produced by these internal plasma currents is given by the Biot-Savart law:

$$A_{\text{in}}(x) = \frac{\mu_0}{4\pi} \int_{\partial V_{\text{in}}} k_{\text{eq}}(x') \frac{1}{|x - x'|} dS' \quad (4.14)$$

Instead of enforcing this strictly, we impose it in weak form using a set of test functions w that are tangent to the interface and divergence-free. By approximating the boundary ∂V_{in} as a thin volumetric shell V_{eq} , we can use the 3D CARIDDI vector basis functions w_k for both the test functions and the equivalent current density expansion.

Expanding the equivalent current as $J_{\text{eq}}(x') = \sum_j I_{\text{eq},j} w_j(x')$ and substituting $\int_{V_{\text{eq}}} J_{\text{eq}} dV = \int_{\partial V_{\text{eq}}} k_{\text{eq}} dS$, the weak form over the equivalent volume becomes:

$$\int_{V_{\text{eq}}} w_k(x) \cdot A_{\text{in}}(x) dV = \sum_j I_{\text{eq},j} \int_{V_{\text{eq}}} w_k(x) \cdot \left[\frac{\mu_0}{4\pi} \int_{V_{\text{eq}}} \frac{w_j(x')}{|x - x'|} dV' \right] dV \quad (4.15)$$

Separating the Vector Potential

In the Tokamak Case: JOEREK evaluates the *total* magnetic vector potential A at the boundary, not just the contribution from the plasma. So we can use linear superposition to separate the external wall sources and internal sources:

$$A = A_{\text{in}} + A_{\text{ext}} \implies A_{\text{in}} = A - A_{\text{ext}} \quad (4.16)$$

Substituting this decoupling into the left-hand side of Equation (4.15) yields:

$$\int_{V_{\text{eq}}} \mathbf{w}_k \cdot (\mathbf{A} - \mathbf{A}_{\text{ext}}) dV = [L_{\text{eq}} I_{\text{eq}}]_k \quad (4.17)$$

The term involving $\mathbf{A}_{\text{ext}}$ represents the projection of the external wall fields onto the equivalent shell, which defines the mutual inductance matrix between the equivalent currents and the wall currents:

$$\int_{V_{\text{eq}}} \mathbf{w}_k \cdot \mathbf{A}_{\text{ext}} dV = [M_{\text{eq},w} I_w]_k \quad (4.18)$$

In the Stellarator case: The magnetic vector potential ansatz $\mathbf{A} = \Psi \nabla \chi$ explicitly represents only the induced field generated by the internal plasma sources. Consequently, it is not necessary to subtract the external wall current contributions. So we only remain with the formulation:

$$\int_{V_{\text{eq}}} \mathbf{w}_k \cdot \mathbf{A}_{\text{in}} dV = [L_{\text{eq}} I_{\text{eq}}]_k \quad (4.19)$$

Stellarator Vector Potential Ansatz and Projection Matrices

We now discretize the magnetic vector potential ansatz using the JOEREK basis functions to derive a projection matrix that maps Ψ onto $\mathbf{A}$.

$$\mathbf{A} = \Psi \nabla \chi \quad (4.20)$$

Applying JOEREK's spatial discretization, the scalar functions Ψ is expanded using the finite element basis functions λ_j :

$$\mathbf{A}(x) = \sum_j \Psi_j \lambda_j(x) \nabla \chi \quad (4.21)$$

We now insert this discretized stellarator representation (4.21) into the remaining integral on the left-hand side of Equation (4.19):

$$\int_{V_{\text{eq}}} \mathbf{w}_k \cdot \mathbf{A} dV = \int_{V_{\text{eq}}} \mathbf{w}_k \cdot \left(\sum_j \Psi_j \lambda_j \nabla \chi \right) dV \quad (4.22)$$

By moving the summation and state variables outside the integrals, we define the geometric projection matrices H_χ :

$$(H_\chi)_{k,j} = \int_{V_{\text{eq}}} \mathbf{w}_k \cdot (\lambda_j \nabla \chi) dV \quad (4.23)$$

This allows us to write the projection of the total plasma vector potential strictly as a matrix-vector product: $H_\chi \Psi$

Inductance Matrices and Equivalent Currents

The coupling between the domains requires the self-inductance matrix L_{eq} connecting the 3D basis functions on the equivalent surface. It is given by the standard volumetric Biot-Savart integral:

$$(L_{\text{eq}})_{k,j} = \int_{V_{\text{eq}}} \mathbf{w}_k(x) \cdot \left(\frac{\mu_0}{4\pi} \int_{V_{\text{eq}}} \frac{\mathbf{w}_j(x')}{|x - x'|} dV' \right) dV \quad (4.24)$$

Reassembling Equation (4.19) with all newly defined matrices gives the stellarator equivalent of the matching condition:

$$H_\chi \Psi = L_{\text{eq}} I_{\text{eq}} \quad (4.25)$$

Inverting the self-inductance matrix L_{eq} yields the explicit equation for the equivalent surface currents required to shield the plasma:

$$I_{\text{eq}} = L_{\text{eq}}^{-1} (H_\chi \Psi) \quad (4.26)$$

Coupling Matrices Transferred to JOEREK

To close the system, CARIDDI must calculate the boundary magnetic field projection $(\mathbf{B} \times \nabla \chi) \cdot \mathbf{n}$ required by JOEREK's weak form and return the linear response matrix. CARIDDI generates a Biot-Savart matrix $B_{\text{eq},\chi}$ that computes the field from the equivalent shell basis functions and projects it along this required direction.

$$\begin{aligned} B &= \sum_j I_{\text{eq},j} \frac{\mu_0}{4\pi} \int_{V_{\text{eq}}} \mathbf{w}_j(x') \times \frac{\mathbf{x}_i - \mathbf{x}'}{|\mathbf{x}_i - \mathbf{x}'|^3} dV' \\ ((\mathbf{B} \times \nabla \chi) \cdot \mathbf{n})_{i,j} &= \sum_j I_{\text{eq},j} \underbrace{\left\{ \left[\frac{\mu_0}{4\pi} \int_{V_{\text{eq}}} \mathbf{w}_j(x') \times \frac{\mathbf{x}_i - \mathbf{x}'}{|\mathbf{x}_i - \mathbf{x}'|^3} dV' \right] \times \nabla \chi(x_i) \cdot \mathbf{n}(x_i) \right\}}_{=(B_{\text{eq},\chi})} \end{aligned} \quad (4.27)$$

In the no-wall limit, external wall currents are entirely absent ($I_w = 0$). Substituting I_{eq} into the boundary field evaluation yields a single dense matrix that is transferred back to JOEREK:

Plasma Boundary Response Matrix (C_χ)

$$C_\chi = B_{\text{eq},\chi} L_{\text{eq}}^{-1} H_\chi \quad (4.28)$$

This matrix maps the unknown JOEREK state vector Ψ directly to its self-consistent contribution to the boundary magnetic field.

So to summarize:

1. H_χ maps JOREK's Ψ to the boundary vector potential.
2. L_{eq}^{-1} converts that vector potential into the physical equivalent surface currents (I_{eq}).
3. $B_{\text{eq},\chi}$ calculates the projected boundary magnetic field resulting from those surface currents.

Outline of the JOREK–CARIDDI Coupling

The coupling workflow currently involves several repeated steps and may appear somewhat redundant. This is part of the ongoing development process and will be streamlined once the implementation is verified to working.

1. Run JOREK (model 180) to generate the `boundary.txt` file.
 2. CARIDDI processes `boundary.txt`, constructs the coupling surface, and computes the corresponding Gauss integration points.
 3. Re-run JOREK (model 180), now using the Gauss points to evaluate $\nabla\chi$ at the specified locations.
 4. CARIDDI runs again, receiving $\nabla\chi$ and using it to assemble the coupling matrices.
 5. For this specific `boundary.txt` and the computed $\nabla\chi$, simulations can now be repeated as needed without redoing the full setup.
-

5 Verification

The preceding chapter derived the stellarator coupling scheme and, as its central result, the plasma boundary response matrix

$$C_\chi = B_{\text{eq},\chi} L_{\text{eq}}^{-1} H_\chi, \quad (5.1)$$

which maps the JOREK boundary flux Ψ to its self-consistent contribution to the boundary field projection $(\mathbf{B} \times \nabla\chi) \cdot \mathbf{n}$. Before this operator is handed to JOREK and used to close a free-boundary simulation, it must be established that the implemented C_χ actually reproduces this defining action. This chapter is dedicated to that verification.

Verifying C_χ in isolation—independently of any JOREK time evolution—is attractive because it separates errors in the response operator itself from errors in the coupled dynamics. The natural way to do so is a *convergence study*: an analytic configuration is chosen for which the exact boundary response is known in closed form, the discrete operator is applied to it, and the residual is monitored as the discretization is systematically refined. A correct implementation must then show the error decreasing at the rate expected from the underlying finite-element approximation, down to the irreducible floor imposed by the finite representation of the boundary data.

The remainder of the chapter is organised accordingly. Section 5.1 introduces the discretization parameters that govern the accuracy of C_χ , explains what each one controls, and defines the representation floor that bounds the achievable error from below. Section 5.2 describes the analytic test case—a current-carrying wire whose fields are known analytically—and the projection scheme that makes it compatible with the $\nabla\chi$ -based representation. Section 5.3 reports the convergence results for two wire orientations, compares the observed order of convergence against the value expected from the discretization, and discusses the sources of the residual error.

5.1 Discretization Parameters and the Representation Floor

The accuracy with which the assembled C_χ reproduces its continuous counterpart is controlled by a set of discretization parameters. They fall into three groups: those that set the resolution of the equivalent shell on which the currents live, those that control the accuracy of the numerical integration, and those that fix the function space in which

the boundary data is represented. A clean convergence study refines the first two while holding the third—and the physical configuration—fixed, so that any reduction in the error is attributable to a genuine improvement of the discrete operator.

Resolution of the equivalent shell

The equivalent surface currents are expanded in the three-dimensional CARIDDI finite-element basis w_k defined on a thin volumetric shell that follows the JOEREK boundary. Two parameters set its resolution:

- `n_elem_tor_J` — the number of elements distributed along the toroidal angle. This is the primary refinement variable used throughout the study, since the non-axisymmetric structure of a stellarator boundary is resolved chiefly in the toroidal direction.
- `nsubi` — the number of poloidal subdivisions of each JOEREK boundary element on the shell. A companion parameter, `nsubi_ctrl`, plays the same role for the auxiliary control surface on which the boundary field is sampled; the code requires `nsubi` to be an integer multiple of `nsubi_ctrl`.

The shell is discretized with trilinear hexahedral elements. The current representation along the toroidal direction is therefore piecewise linear, which fixes the convergence rate one may expect from refining `n_elem_tor_J`: for a smooth field, a piecewise-linear approximation converges with second order in the element size, i.e. as $\mathcal{O}(N_{\text{elem,tor}}^{-2})$.

Numerical integration

All coupling matrices are assembled from volume and surface integrals (Biot–Savart kernels, mass-matrix products, and Fourier projections), each evaluated by Gaussian quadrature. The relevant parameters are the poloidal and toroidal quadrature orders on the shell (`n_gauss_pol_J`, `n_gauss_tor_J`), the quadrature orders entering the self-inductance L_{eq} (`ng_pol_Leq`, `ng_tor_Leq`), and the number of toroidal planes (`ntor_ctrl`) on which $B_{\text{eq},\chi}$ performs its Fourier projection. These parameters do not change *what* is being computed, only how accurately the integrals are evaluated. The toroidal sampling is the most delicate: the integrands contain products of the basis functions with $\nabla\chi$, so their toroidal spectrum extends to roughly twice the highest retained harmonic. If the toroidal quadrature or the number of control planes falls below the corresponding Nyquist limit, the high-harmonic blocks of H_χ and $B_{\text{eq},\chi}$ are corrupted by aliasing. The toroidal sampling must therefore comfortably exceed the spectral content of these products.

In addition, the construction of the analytic reference itself uses a finite-difference step $\text{DELTA_FD} = 10^{-5} \text{ m}$ to evaluate the gradients of $\nabla\chi$ by central differences (see Section 5.2); and the control surface is offset from the current shell by a small distance d_shift to keep the Biot–Savart kernels non-singular. Both introduce a small, resolution-independent bias.

Function space and the representation floor

The boundary flux Ψ is represented in the JOEREK basis as a tensor product of a poloidal C^1 cubic Bézier (Hermite) basis and a toroidal Fourier basis. The toroidal content is fixed by the set of retained harmonics, specified through `ntori` (their number) and `nharmi` (their mode numbers). Unlike the parameters above, these define the function space itself: they change *what* the test can represent, not how accurately a given representation is computed.

This finite function space imposes a hard lower bound on the achievable verification error. Even an exact C_χ cannot produce a boundary response that is more accurate than the JOEREK basis can represent. Two diagnostics quantify this floor directly by measuring how well the analytic input and the analytic target survive a projection onto the JOEREK basis and back:

- **Diagnostic 1** (Ψ floor): the compatible flux ψ_χ is L^2 -projected onto the JOEREK basis and reconstructed at the integration points; the relative discrepancy $\|\psi_\chi - \psi_\chi^{\text{proj}}\| / \|\psi_\chi\|$ measures how cleanly the *input* is represented.
- **Diagnostic 2** (f_{RHS} floor): the same round-trip is applied to the target boundary quantity $f_{\text{RHS}} = (\mathbf{B}_\chi \times \nabla\chi) \cdot \mathbf{n}$. Because this is precisely the quantity C_χ must reproduce, its representation error is the irreducible floor below which the verification error cannot fall, regardless of how finely the shell is resolved.

A meaningful convergence study therefore drives the discretization error down toward—but not below—the Diagnostic 2 floor; once the two become comparable, the test has become limited by the function space rather than by C_χ , and further mesh refinement yields no improvement.

5.2 Test Case: Analytical Wire

The boundary response matrix C_χ maps the scalar flux Ψ on the boundary to the self-consistent representation of the boundary magnetic field,

$$C_\chi \Psi = (\mathbf{B} \times \nabla\chi) \cdot \mathbf{n}. \quad (5.2)$$

To verify this action we use a configuration whose magnetic field and vector potential are known in closed form: a circular current-carrying wire. The strategy is to evaluate the analytic field at the boundary integration points, form the two sides of (5.2) independently—the left-hand side by applying the assembled matrix to the analytic flux, the right-hand side directly from the analytic field—and compare them.

Compatibility of the analytic field with the representation

A subtlety must be addressed first. The matrix C_χ is assembled for one specific vacuum field $\nabla\chi$ and the ansatz $\mathbf{A} = \Psi\nabla\chi$. It can therefore not be applied to an arbitrary analytic field that was not constructed in this representation. To preserve the analytic reference, we extract the component of the analytic vector potential $\mathbf{A}_{\text{total}}$ that is aligned with $\nabla\chi$. Decomposing into an aligned part and a perpendicular remainder ($\mathbf{A}_{\text{rest}} \cdot \nabla\chi = 0$),

$$\begin{aligned}\mathbf{A}_{\text{total}} &= \mathbf{A}_\chi + \mathbf{A}_{\text{rest}} \\ &= \psi_\chi \nabla\chi + \mathbf{A}_{\text{rest}},\end{aligned}\tag{5.3}$$

and taking the dot product with $\nabla\chi$ yields

$$\mathbf{A}_{\text{total}} \cdot \nabla\chi = \psi_\chi |\nabla\chi|^2,\tag{5.4}$$

so that the compatible scalar flux is

$$\psi_\chi = \frac{\mathbf{A}_{\text{total}} \cdot \nabla\chi}{|\nabla\chi|^2}.\tag{5.5}$$

The corresponding magnetic field follows from the curl of the aligned part,

$$\begin{aligned}\mathbf{B}_\chi &= \nabla \times \mathbf{A}_\chi = \nabla \times (\psi_\chi \nabla\chi) \\ &= \nabla\psi_\chi \times \nabla\chi + \psi_\chi \underbrace{(\nabla \times \nabla\chi)}_{=0} = \nabla\psi_\chi \times \nabla\chi.\end{aligned}\tag{5.6}$$

Having isolated the components ψ_χ and $\mathbf{B}_\chi$ of the analytic solution that live in the $\nabla\chi$ representation, the matrix can be tested through the consistent identity

$$C_\chi \psi_\chi = (\mathbf{B}_\chi \times \nabla\chi) \cdot \mathbf{n}.\tag{5.7}$$

Wire configuration

The analytic field is that of a single circular current loop carrying a unit current, placed well inside the boundary at the centroid of the integration-point cloud so that it stays clear of the shell. The loop is tilted by an angle α about the x -axis. The tilt controls the toroidal spectrum of the test field and is the lever used to probe different parts of C_χ :

- For $\alpha = 0^\circ$ the loop is coaxial with the symmetry axis and its field is axisymmetric. The compatible flux ψ_χ is nevertheless not purely $n = 0$: the division by $|\nabla\chi|^2$ in (5.5) mixes in the toroidal content of $\nabla\chi$, which for the five-period stellarator used here carries the harmonics $n = 0, 5, 10, \dots$. The case thus exercises the curved boundary geometry and the full quadrature chain, not merely the axisymmetric block of C_χ .
- For $\alpha = 1^\circ$ the rigid tilt breaks axisymmetry and adds genuine non-axisymmetric structure to the field, increasing the amount of information the discrete operator must reproduce.

Numerical treatment

Evaluating (5.7) at the boundary integration points requires three ingredients that are computed numerically and were the subject of dedicated care:

- *Gradient of the compatible flux.* The field $\mathbf{B}_\chi$ in (5.6) needs $\nabla\psi_\chi$, and because ψ_χ depends on $\nabla\chi$ through (5.5), its gradient is obtained by central finite differences: $\nabla\chi$ is evaluated at the integration points shifted by $\pm\delta$ along each Cartesian direction (with $\delta = \text{DELTA_FD} = 10^{-5}$ m) and differenced. This avoids the noisy gradients that a direct differentiation of the discrete $\nabla\chi$ would produce.
- *Projection onto the JOREK basis.* The analytic quantities ψ_χ and $f_{\text{RHS}} = (\mathbf{B}_\chi \times \nabla\chi) \cdot \mathbf{n}$ are constructed in real space at the Gauss points and must be expressed in the JOREK Bézier–Fourier basis before C_χ can act on them. This is done by an L^2 projection: the discrete mass matrix of the basis is assembled on the integration measure and solved against the analytic data, yielding the coefficient vectors entering (5.7).
- *Boundary normal.* The outward unit normal $\mathbf{n}$ at each integration point is built from the tangential derivatives of the boundary surface, including the full toroidal derivative, using the same routine that constructs the control surface. Its global orientation is fixed once by requiring the normal to point outward at the most outboard sample point.

Verification procedure

With these ingredients the test proceeds as follows:

1. Evaluate the analytic $\mathbf{A}$ and $\mathbf{B}$ of the (optionally tilted) wire at the boundary integration points, and extract ψ_χ from (5.5).

2. Build B_χ from (5.6) using the finite-difference gradient, and assemble the target boundary quantity f_{RHS} .
3. L^2 -project ψ_χ and f_{RHS} onto the JOEREK basis.
4. Apply the assembled matrix, $C_\chi \psi_\chi$, and compare it against f_{RHS} in the sense of (5.7).

The discrepancy between the two sides, reconstructed in real space at the integration points and normalised by the magnitude of f_{RHS} , is the error reported in the next section.

5.3 Convergence Results

The convergence study fixes the wire configuration and the JOEREK function space and refines the toroidal shell resolution $n_{\text{elem_tor_J}}$ through the sequence $N_{\text{elem,tor}} \in \{24, 48, 96, 192, 384\}$, for the two wire tilts $\alpha = 0^\circ$ and $\alpha = 1^\circ$. The reported error is the real-space round-trip discrepancy between $C_\chi \psi_\chi$ and f_{RHS} , normalised by $\|f_{\text{RHS}}\|$. From two successive refinements the observed order of convergence is estimated as

$$p = \frac{\ln(e_N/e_{2N})}{\ln 2}, \quad (5.8)$$

where e_N is the error at resolution N . Because every companion parameter—the quadrature orders, the finite-difference step, and above all the retained harmonic set—is held fixed during the sweep, the representation floor (Diagnostic 2) is constant and provides a fixed reference for the achievable accuracy. Its value, together with the input floor (Diagnostic 1), is 0.049 % and 0.21 % for $\alpha = 0^\circ$, and 0.076 % and 0.76 % for $\alpha = 1^\circ$.

The results are listed in Table 5.1 and shown on logarithmic axes in Figure 5.1.

Expected and observed order of convergence

As established in Section 5.1, the equivalent shell is built from trilinear hexahedral elements, so the toroidal representation of the currents and the boundary geometry is piecewise linear. For a smooth field this leads to second-order convergence in the element size, and—if the toroidal resolution were the only source of error—one would expect $e \sim \mathcal{O}(N_{\text{elem,tor}}^{-2})$. The coarse-grid behaviour is consistent with this: the first refinement of the axisymmetric case yields an order of $p \approx 1.8$, close to the theoretical value, and both curves in Figure 5.1 track the $\mathcal{O}(N^{-2})$ reference slope over the first decade of refinement.

Table 5.1: Convergence of the C_χ verification error under refinement of the toroidal shell resolution $N_{\text{elem,tor}}$ for the untilted ($\alpha = 0^\circ$) and tilted ($\alpha = 1^\circ$) wire. The error is the real-space round-trip residual of (5.7); the order p is estimated from (5.8). The representation floor (Diagnostic 2) is 0.21 % and 0.76 %, respectively.

$N_{\text{elem,tor}}$	$\alpha = 0^\circ$		$\alpha = 1^\circ$	
	error (%)	order p	error (%)	order p
24	27.579	—	30.790	—
48	7.853	1.81	13.006	1.24
96	2.639	1.57	4.204	1.63
192	1.280	1.04	2.086	1.01
384	0.984	0.38	1.590	0.39

The observed order does not stay at two, however; it decays steadily as the mesh is refined, reaching $p \approx 0.4$ at the finest resolution for both tilts. This is the expected consequence of refining a single parameter while all others are held fixed. Each frozen parameter contributes an error that does not diminish with $N_{\text{elem,tor}}$, the dominant one being the finite harmonic truncation of the JOREK basis quantified by the Diagnostic 2 floor; the fixed toroidal quadrature, the finite-difference step, and the control-surface offset add further constant contributions. The total error is well described by

$$e(N) \approx C N^{-2} + e_{\text{floor}}, \quad (5.9)$$

in which the discretization term decays at the expected second-order rate while the floor term does not. Substituting (5.9) into the order estimate (5.8) shows that the measured p must drift toward zero once $C N^{-2}$ becomes comparable to e_{floor} —precisely the behaviour seen in the data. Accordingly, the error plateaus a few times above the Diagnostic 2 floor (0.98 % against 0.21 % for $\alpha = 0^\circ$, 1.59 % against 0.76 % for $\alpha = 1^\circ$) rather than vanishing.

The two tilts behave alike. The tilted case carries genuine non-axisymmetric content and therefore starts from a larger error and a higher representation floor, but its convergence follows the same pattern; the non-monotonic order at the coarsest grids reflects the under-resolution of that non-axisymmetric content, which is cleared once the toroidal sampling is sufficient. It is worth stressing that the untilted case is not a trivial axisymmetric check: as discussed in Section 5.2, the $\nabla\chi$ of the five-period stellarator spreads ψ_χ across the harmonics $n = 0, 5, 10, \dots$, and the curved torus is approximated by linear elements, so the $\alpha = 0^\circ$ sweep is a genuine test of the geometric and quadrature convergence of C_χ .

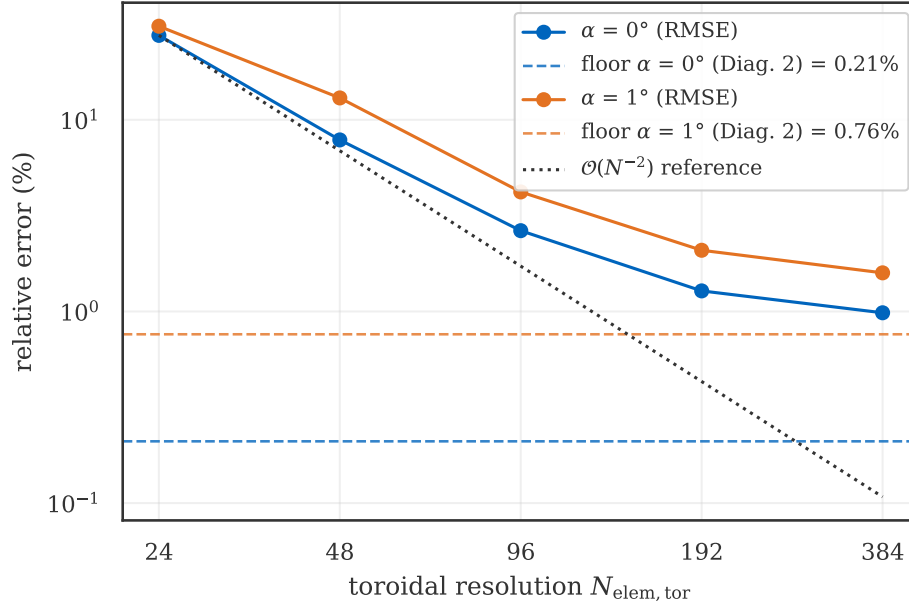


Figure 5.1: Convergence of the C_χ verification error with the toroidal shell resolution, for $\alpha = 0^\circ$ and $\alpha = 1^\circ$. The dashed lines mark the representation floor (Diagnostic 2) of each case; the dotted line is the $\mathcal{O}(N^{-2})$ slope expected from the trilinear hexahedral shell elements. Both curves follow the expected slope at coarse resolution and bend over as they approach their floor.

Taken together, the study confirms that the implemented C_χ reproduces its defining boundary action: the error decreases at the order expected from the shell discretization until it becomes limited by the finite JOEREK representation, at which point further mesh refinement cannot help. Reducing the residual below the present plateau requires lowering the floor itself—chiefly by enriching the retained harmonic set—rather than by refining the toroidal mesh alone.

Behaviour of the remaining discretization parameters

[Placeholder — the convergence behaviour of the remaining discretization parameters introduced in Section 5.1 ($n_{\text{sub}i}$, the Gaussian quadrature orders, the finite-difference step DELTA_FD , and the retained harmonic set) will be reported here once the corresponding sweeps are available.]

Nomenclature

The following symbols are used throughout this thesis:

Symbol	Description
B	Magnetic field strength

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